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22 Sep. 2026 Lecture 5 Logic Synthesis: 2-Level Logic: Basics

Try to find minimal SOP logic: many AND gates, 1 OR gate
fewest AND gates, one with fewest input wires

Strategy:

- 1: Don't try for the best, just a good answer
- 2: Iterative improvement.

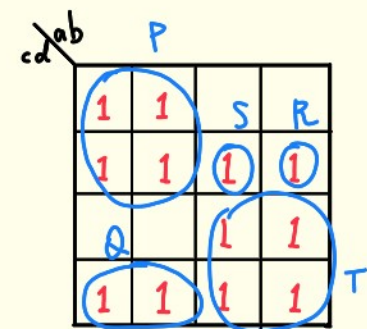
Famous result from 1950s: best solution is composed of cover of primes. Every cover is as big as possible

Both solutions are "irredundant" (compare best and relatively good solution)

23 Sep. 2026 The reduced-Expand-Irredundant-Loop

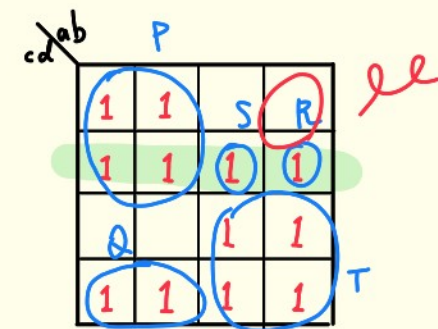
Famous heuristic method for 2-level logic optimization

Starting from a relatively bad Karnaugh map:



abcd	F	cube-label
0*0*	1	P
0*10	1	Q
1001	1	R
1101	1	S
1*1*	1	T

Next step: Expand each cube to be prime
make each cube as big as possible. There might be different ways to do so.



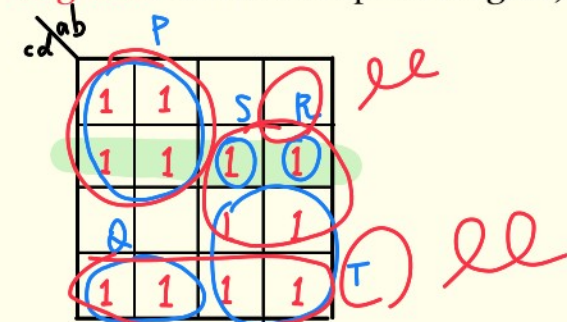
Next: remove redundant cubes: heuristic

abcd	F	cube-label
0*0*	1	P
0*10	1	Q
1001	1	R
1101	1	S
1*1*	1	T

make it "minima"

Reduce: Reduce the prime cover
Take each cube, "shrink it", but do not uncover any 1s
this is not necessarily a prime cover

Big idea: when we expand it again, maybe we get a new, better solution



abcd	F	cube-label
0*0*	1	P
0*10	1	Q
1001	1	R
1101	1	S
1*1*	1	T

This will not generally happen!

Famous tool: **ESPRESSO** for 2-level minimization
started at IBM, finished at Berkeley

Detailed introduction to “EXPAND”

core representation: PCN cube list

What does "expand a cube" mean? **Remove** variables from cube

Expand: Transform into a Covering Problem

Given matrix of R rows * C columns. Matrix has only 1s & 0s in it. only draw the 1s

Choose smallest rows so that using these rows every column has at least a 1 in it.

Very good heuristics to get decent fast solutions for these

Expand -> a Covering Problem on the Blocking Matrix

build a cube of the 0s in F (called the OFF set)

(we need to know what our cubes cannot touch when it expands)

How we do this? URP Complement of the starting cover of the function

EXPRESSO: Collection of Elegant Heuristics